

Assignment 1 - Real-World Banking ETL

Background

You are an analytics engineer at a bank. Each month, a third-party agency provides a file containing updated vehicle information – including model names, energy classes (eg., electric, gas, diesel), and other specifications.

Currently, a business analyst manually runs queries to calculate estimated profits per vehicle and to generate new auto loan offers. These calculations are based on:

- The energy class of the car (e.g., electric, gasoline, diesel)
- The floating interest rate, which is updated monthly
- The bank’s margin that is added on top of the base rate

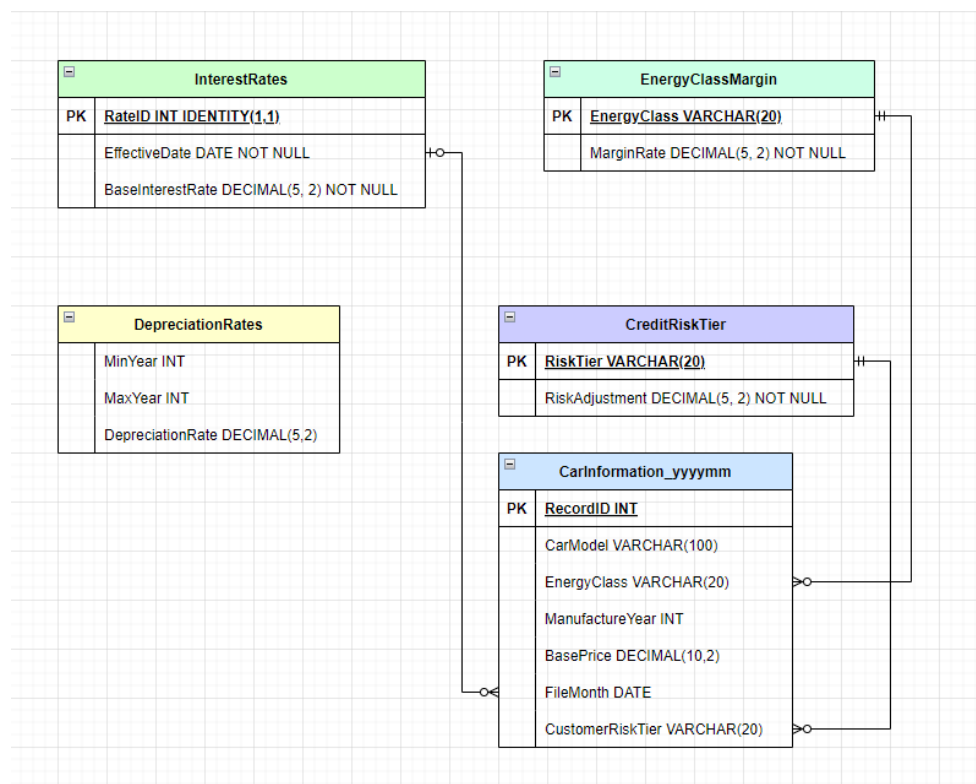
Requirements

You are required to build a SQL-based ETL job in SSMS (SQL Server Management Studio) that automatically:

- Ingests the monthly vehicle data into a staging table.
- Validates and transforms the data (e.g., corrects data types and formats).
- Loads the enriched data into a final target table. The business analyst should be able to simply run “SELECT * FROM LoanProfitEstimates_yyyyymm;” to get the final results each month.

Relationship

The diagram below illustrates the relationships between the key tables used in the query:



Grading Criteria

Grade G

To achieve a G, your solution must:

- Dynamically select the correct CarInformation_yyyymm table based on the current month.
- If the base interest rate is missing for the current month, fall back to the most recent available month.
- Generate the output as a new results table named LoanProfitEstimates_yyyymm, with no errors.

Grade VG

To achieve a VG, your solution must fulfill all G-level requirements, plus:

- Include clear and well-structured code, with explanatory comments and efficient performance.
- Implement logging to capture errors with descriptive messages, such as:
 - "Base interest rate for the current month is missing. Value from the previous month was used."
 - "CarInformation_yyyymm table for the current month not found."
- Send a confirmation email after execution, stating whether the job/jobs completed successfully and including details such as the number of rows processed and the name of the generated output table.